

Ticket Media & Credential Management

Functional Design & Backend Screens

TICVAI | Professional Reference Document

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Area 15 — Ticket Media & Credential Management

Board 1 — Virtual Ticket & Multi-Media Architecture

10 Backend Screens — Detailed Scope of Work

The screenshot displays the TICVAI dashboard for Area 15 - Ticket Media & Credential Management. The interface is divided into several sections:

- 15.1.1 Virtual Ticket Command Center:** Shows key metrics like New Virtual Tickets (126,842), Active (98,672), Pending Activation (7,312), Used/Consumed (14,128), Suspended (1,842), and Expired (3,288). It also includes a 'Recent Exceptions' section and a 'Top Media Distribution' pie chart.
- 15.1.2 Virtual Ticket Identity & Master Record Configuration:** Provides fields for Virtual Ticket Identity, Master Record, and various configuration options like Product, Event, and Ticket Holder.
- 15.1.3 Virtual Ticket Status & Lifecycle Model:** Displays a lifecycle flowchart and a table of transition rules (e.g., Created, Pending, Active, Suspended, Expired).
- 15.1.4 Media Type & Credential Technology Registry:** A table listing various media types (Dynamic QR, Barcode, NFC, etc.) and their associated technologies.
- 15.1.5 Multi-Media Binding & Association Rules:** Shows rule summaries and configuration options for different media types.
- 15.1.6 Credential Identity, Token & Reference Mapping:** Displays mapping details for credentials and tokens.
- 15.1.7 Entitlement & Cross-Media Synchronization Rules:** Shows synchronization events and rules for entitlements.
- 15.1.8 Media Activation, Priority & Fallback Rules:** Configures activation settings, priorities, and fallback rules for different media types.
- 15.1.9 Media Replacement, Revocation & Rebinding Rules:** Shows replacement policies and rules for media management.
- 15.1.10 Virtual Ticket Architecture Testing, Governance & Audit:** Includes a security simulator, test results, and an audit trail for system operations.

Board Objective

Board 1 establishes the **core credential architecture of TICVAI**.

The most important principle is:

The Virtual Ticket is the ticket. QR, barcode, RFID, NFC, Apple Wallet, Google Wallet, face recognition, wristbands, cards and other credentials are media linked to that Virtual Ticket.

TICVAI must therefore separate the **ticket identity and entitlement** from the **media used to identify/present it**.

The architecture is:

Order / Entitlement



Virtual Ticket ID — Authoritative Master Record



0..N Credential / Media Bindings



QR | Barcode | RFID | NFC | Apple Wallet | Google Wallet | Face | Card | Wristband | Future Media

A single Virtual Ticket may have **multiple active media simultaneously**, but all of them resolve to the same authoritative ticket and entitlement state.

15.1.1 — Virtual Ticket Command Center

Purpose

Provide administrators and operations teams with a centralized view of all Virtual Tickets and their associated media across TICVAI.

This is the primary administrative entry point into the Virtual Ticket architecture.

Executive KPIs

Display:

- Total Virtual Tickets
- Active
- Pending Activation
- Suspended
- Used / Consumed
- Partially Consumed
- Expired
- Cancelled
- Revoked
- Virtual Tickets with Multiple Media
- Virtual Tickets with No Active Media
- Media Binding Exceptions
- Credential Synchronization Issues

Virtual Ticket Directory

Each record should show:

- Virtual Ticket ID
- Product
- Event / Performance
- Ticket Holder
- Order Reference
- Ticket Type
- Seat / Resource where applicable
- Valid From / To
- Ticket Status
- Usage Status
- Number of Linked Media
- Primary Media
- Last Credential Activity

- Last Modified

Example:

VT-2026-009821

Product: VIP Concert

Customer: Ahmed Hassan

Seat: A-18

Status: Active

Media: 4

→ Dynamic QR — Active

→ RFID — Active

→ Apple Wallet — Active

→ Face — Active

Filters

Filter by:

- Brand
- Venue
- Event
- Product
- Performance
- Ticket Type
- Channel
- Customer
- Virtual Ticket Status
- Media Type
- Number of Media
- Validity
- Usage Status

Quick Actions

Authorized users may:

- View Virtual Ticket
- View Media
- Bind Media
- Replace Media
- Suspend
- Reactivate
- Revoke Media
- View Usage
- View Audit
- Diagnose Credential

AI Capability

AI can identify unusual credential conditions.

Example:

“128 Virtual Tickets for tomorrow's event have no active customer-presentable media.”

or:

“17 Virtual Tickets have multiple credentials with inconsistent activation states.”

AI is advisory; it does not change ticket validity.

Acceptance Condition

Authorized users can locate any Virtual Ticket and immediately understand its ticket status, entitlement, usage state and all associated media from one centralized workspace.

15.1.2 — Virtual Ticket Identity & Master Record Configuration

Purpose

Define the authoritative Virtual Ticket object used throughout TICVAI.

This screen is extremely important because **the Virtual Ticket—not the QR/Rfid/card—is the master ticket record.**

Virtual Ticket Master Identity

Every issued ticket must receive a unique persistent:

Virtual Ticket ID

Example:

VT-2026-000009821

The identifier must remain persistent regardless of:

- Media replacement
- QR regeneration
- RFID replacement
- Wallet updates
- Device changes
- Face enrollment changes
- Ticket reprint

Master Record References

The Virtual Ticket should reference, where applicable:

- Order
- Order Line
- Reservation
- Product
- Product Version
- Event
- Performance / Timeslot
- Venue
- Ticket Holder
- Participant
- Customer
- Membership
- Seat

- Resource
- Price Snapshot
- Entitlements
- Validity
- Access entitlement
- Fulfillment status

Virtual Ticket Configuration

Configure:

- ID generation pattern
- Ticket classification
- Ticket ownership model
- Holder assignment requirements
- Transferability reference
- Validity model
- Consumption model
- Entitlement model
- Media requirements

Media Independence

A mandatory architectural rule:

Changing, replacing or revoking a credential medium must not require replacement of the underlying Virtual Ticket unless the business transaction itself requires a new ticket.

Example:

RFID wristband lost.

Old:

RFID-77812 → VT-009821

Revoke RFID.

New:

RFID-99142 → VT-009821

The Virtual Ticket remains unchanged.

Source of Truth

The Virtual Ticket owns/references authoritative ticket-level state such as:

- Valid
- Suspended
- Cancelled
- Expired
- Used
- Partially Used
- Remaining Entitlements

Media must not independently maintain conflicting entitlement states.

Acceptance Condition

Every issued ticket has one persistent, media-independent Virtual Ticket identity that remains authoritative regardless of which credential technologies are attached to it.

15.1.3 — Virtual Ticket Status & Lifecycle Model

Purpose

Configure the standardized lifecycle of a Virtual Ticket independently from the lifecycle of individual media.

Recommended Ticket States

Support:

Created → Pending Fulfillment → Active → Partially Used → Used / Consumed → Expired

with governed alternative states:

- Suspended
- Cancelled
- Voided
- Reissued / Superseded
- Refunded
- Transferred
- Blocked

Important Separation

TICVAI must distinguish:

Virtual Ticket Status

from:

Credential Media Status

For example:

Virtual Ticket:

ACTIVE

Media:

- QR — Active
- RFID — Lost / Revoked
- Apple Wallet — Active
- Face — Active

The ticket itself remains valid.

Transition Rules

Configure which transitions are allowed.

Example:

Active → Suspended ✓

Suspended → Active ✓

Active → Cancelled ✓

Used → Active × without authorized exception

Trigger Sources

Lifecycle changes may originate from:

- Order Management
- Cancellation
- Refund
- Upgrade / Conversion
- Ticket Transfer
- Membership
- Expiry
- Access usage
- Authorized operator
- API
- Scheduled process

Cross-Media Propagation

A Virtual Ticket status change must propagate appropriately to all media.

Example:

Virtual Ticket = Cancelled

Then:

QR → unusable

RFID → unusable

Wallet → update/revoke as technically supported

Face credential association → inactive for this entitlement

Acceptance Condition

TICVAI maintains one governed Virtual Ticket lifecycle and consistently reflects relevant state changes across every credential associated with that ticket.

15.1.4 — Media Type & Credential Technology Registry

Purpose

Maintain the centralized catalogue of credential technologies supported by TICVAI.

This makes the credential architecture extensible rather than hard-coded.

Supported Media Categories

The registry should support, at minimum:

Digital

- QR
- Dynamic QR
- Barcode
- Mobile Ticket
- PDF
- Apple Wallet
- Google Wallet

Physical

- RFID Card
- RFID Wristband

- NFC Card
- NFC Wristband
- Printed Ticket
- Thermal Ticket
- Membership Card
- Custom Wearable

Biometric

- Face Recognition Reference

Future

- Tenant-defined / integration-based credential technology

Media Profile

For each technology configure:

- Media Type ID
- Name
- Category
- Provider
- Technology
- Token format
- Generation method
- Validation mechanism
- Supports visual design
- Supports dynamic update
- Supports revocation
- Supports expiration
- Supports offline reference
- Supports replacement
- Supports encryption/signing
- Supported channels
- Supported devices
- Integration adapter

Provider Independence

For example, TICVAI may support several RFID providers.

Therefore:

Media Type ≠ Vendor

RFID is the technology/media type.

Different hardware/provider integrations can implement it.

Extensibility

New credential technologies should be introduced through configurable/adaptor-based architecture without redesigning the Virtual Ticket core.

Acceptance Condition

TICVAI maintains a governed, extensible registry of all supported media and credential technologies with their capabilities, providers and technical behavior.

15.1.5 — Multi-Media Binding & Association Rules

Purpose

Configure how **one Virtual Ticket can be associated with multiple media simultaneously**.

This is one of the most important screens in Area 15.

Relationship Model

TICVAI must support:

1 Virtual Ticket → 0..N Credential Media

Example:

VT-009821

- Dynamic QR
- RFID Wristband
- Apple Wallet
- Face Recognition

Binding Configuration

Configure:

- Allowed media
- Mandatory media
- Optional media
- Maximum active media
- Minimum media required
- Primary media
- Secondary media
- Backup media
- Temporary media
- Media combination
- Simultaneous activation
- Exclusive activation

Media Combination Rules

Examples:

Annual Pass

Required:

- Face

Optional:

- Mobile QR
- RFID Card

Waterpark

Required:

- RFID Wristband

Optional:

- Mobile QR before wristband collection

Concert

Default:

- Dynamic QR

Optional:

- Apple Wallet
- Google Wallet

Mutual Exclusivity

Allow rules such as:

Physical RFID activated → Temporary paper credential automatically revoked.

Or:

Face + RFID may remain simultaneously active.

Binding Scope

Rules may depend on:

- Product
- Ticket Type
- Event
- Venue
- Customer Type
- Membership
- Channel
- Age
- Country
- Access environment

Acceptance Condition

TICVAI can govern which and how many media may simultaneously represent one Virtual Ticket without creating duplicate ticket entitlements.

15.1.6 — Credential Identity, Token & Reference Mapping

Purpose

Define how individual media identifiers resolve securely back to the authoritative Virtual Ticket.

Credential Resolver Concept

TICVAI should implement a centralized **Credential Resolver Service**.

Input could be:

QR Token →

RFID UID →

Wallet Object →

NFC Token →

Face Reference →

and the resolver returns:

Virtual Ticket ID

Mapping Record

Store/reference:

- Credential Binding ID
- Virtual Ticket ID
- Media Type
- Credential Reference
- Token / Identifier
- Provider Reference
- Issued Date
- Activation Date
- Expiry
- Status
- Version
- Security profile

Security

Sensitive credential values should not unnecessarily be exposed in administrative UI or logs.

Support appropriate:

- Tokenization
- Hashing
- Encryption
- Signed payloads
- Key references
- Masking

depending on credential technology.

Biometric Principle

For face recognition, TICVAI should normally reference an authorized biometric identity/template held in the appropriate biometric service rather than embedding raw biometric data directly into the Virtual Ticket.

Resolver API

Conceptually:

Resolve Credential

Input:

Media Type = RFID

Credential = 04:A2:...

Output:

Virtual Ticket = VT-009821

Acceptance Condition

Every supported credential can securely and consistently resolve to its authoritative Virtual Ticket without duplicating ticket entitlement data within individual media.

15.1.7 — Entitlement & Cross-Media Synchronization Rules

Purpose

Ensure all media attached to a Virtual Ticket share the same authoritative ticket and entitlement state.

Critical Principle

Media does **not** own independent entitlement.

For example, TICVAI must prevent:

QR = Used

RFID = Unused

Face = Unused

from being interpreted as three independent admission entitlements.

Instead:

Virtual Ticket Usage = Used

and all media resolve to that same state.

Synchronization Events

Configure behavior following:

- Entry
- Exit
- Redemption
- Partial consumption
- Cancellation
- Refund
- Suspension
- Reactivation
- Transfer
- Upgrade
- Expiry
- Replacement

Example

Customer enters using face recognition.

Face Credential

↓

Virtual Ticket resolved

↓

Access transaction recorded

↓

Virtual Ticket usage updated

↓

QR/RFID/Wallet subsequently see the updated ticket state.

Partial Entitlements

Support tickets containing multiple consumable rights.

Example:

Theme Park Annual Pass:

- Main admission
- 2 Guest Admissions
- 5 Parking Uses
- 10% F&B Benefit

Credential media all resolve to the same remaining entitlements.

Synchronization Failure**Detect:**

- Delayed updates
- Conflicting states
- Offline transactions pending synchronization
- Provider update failures
- Stale wallet credentials

Acceptance Condition

All credential media consistently consume and represent the same authoritative Virtual Ticket entitlement and usage state.

15.1.8 — Media Activation, Priority & Fallback Rules**Purpose**

Configure when each credential becomes active and how alternative media behave if the preferred credential cannot be used.

Activation Methods**Support:**

- Immediate on issuance
- On ticket activation
- On download
- On wallet installation
- On RFID assignment
- On face enrollment
- On first use
- On event date
- Manual activation
- Scheduled activation

Credential Priority**Configure:****Primary**

Face Recognition

Secondary

RFID

Fallback

Dynamic QR

This does not mean they represent different tickets—they all resolve to the same Virtual Ticket.

Fallback Scenarios

Examples:

Face recognition unavailable
→ use RFID.

RFID lost
→ use Mobile QR.

Customer phone unavailable
→ issue temporary physical credential.

Temporary Credentials

Configure:

- Temporary media
- Validity duration
- One-time use
- Automatic expiration
- Replacement behavior
- Original-media impact

Acceptance Condition

TICVAI can activate and prioritize multiple credential media and provide controlled fallback options without compromising the underlying Virtual Ticket.

15.1.9 — Media Replacement, Revocation & Rebinding Rules

Purpose

Configure controlled handling of lost, stolen, damaged, compromised or replaced credential media.

Scenarios

Support:

- Lost RFID card
- Damaged wristband
- Compromised QR
- New mobile device
- Wallet credential replacement
- Face re-enrollment
- Printed ticket replacement
- Incorrect credential assignment

Core Principle

Replacing media should normally **not replace the Virtual Ticket**.

Example:

VT-009821

Old RFID:

RFID-88721 — REVOKED

New RFID:

RFID-99211 — ACTIVE

Virtual Ticket:

VT-009821 — ACTIVE

Replacement Rules**Configure:**

- Reissue allowed
- Number of replacements
- Replacement fee reference
- Approval required
- Identity verification
- Old media automatically revoked
- Grace period
- Simultaneous media policy
- Reason mandatory
- Supervisor approval

Lost/Stolen**Allow immediate:**

- Suspend Media
- Revoke Media
- Replace Media

without unnecessarily cancelling the underlying ticket.

Rebinding

Authorized users may unbind/rebind credentials subject to governance.

Every change records:

- Old binding
- New binding
- User
- Date/time
- Reason
- Approval
- Related transaction

Acceptance Condition

Credential media can be securely replaced, revoked and rebound while maintaining continuity of the underlying Virtual Ticket and a complete audit history.

15.1.10 — Virtual Ticket Architecture Testing, Governance & Audit

Purpose

Provide the final testing and governance environment for Virtual Ticket and multi-media configurations.

Configuration Simulator

Administrators should test scenarios such as:

Scenario 1

VIP Ticket

→ QR + Apple Wallet

Scenario 2

Annual Pass

→ Face + RFID + Mobile QR

Scenario 3

Waterpark

→ Mobile QR before arrival

→ RFID wristband after check-in

Lifecycle Simulation

Test:

Virtual Ticket Created

↓

QR Generated

↓

RFID Bound

↓

Face Enrolled

↓

Entry via Face

↓

Ticket State Updated

↓

RFID Presented

↓

Same authoritative entitlement evaluated.

Failure Testing

Simulate:

- Lost RFID
- Revoked QR
- Expired wallet token
- Face unavailable
- Offline device
- Duplicate credential presentation
- Synchronization failure
- Provider outage
- Credential binding conflict

Architecture Validation

Automatically check for:

- Media without resolver configuration
- Invalid media combinations
- Missing security profile
- Missing fallback

- Conflicting activation rules
- Excessive active credentials
- Broken provider integration
- Invalid lifecycle dependencies

Approval Workflow

Recommended:

Configuration Owner
 → Technical Review
 → Security Review where applicable
 → Operations Review
 → Approval
 → Publication

Audit

Record:

- Configuration changes
- Media bindings
- Rebindings
- Activation
- Suspension
- Revocation
- Replacement
- Resolver changes
- Rule changes
- Approvals
- Actor
- Timestamp
- Before/After

AI Configuration Review

AI may identify risks such as:

“RFID replacement currently leaves the old RFID active for 30 minutes. This conflicts with the configured lost-credential security policy.”

AI may recommend but should not independently change security policies.

Acceptance Condition

TICVAI can test and govern Virtual Ticket and multi-media configurations before production and retain complete traceability of configuration and credential-binding changes.

Board 1 — Final Screen Register

Screen n	Backend Screen	Primary Responsibility
15.1.1	Virtual Ticket Command Center	Central Virtual Ticket management

Screen	Backend Screen	Primary Responsibility
15.1.2	Virtual Ticket Identity & Master Record Configuration	Authoritative ticket identity
15.1.3	Virtual Ticket Status & Lifecycle Model	Master ticket lifecycle
15.1.4	Media Type & Credential Technology Registry	Supported credential technologies
15.1.5	Multi-Media Binding & Association Rules	One ticket → many media
15.1.6	Credential Identity, Token & Reference Mapping	Credential resolver
15.1.7	Entitlement & Cross-Media Synchronization Rules	Shared entitlement/usage
15.1.8	Media Activation, Priority & Fallback Rules	Activation and fallback
15.1.9	Media Replacement, Revocation & Rebinding Rules	Credential continuity
15.1.10	Virtual Ticket Architecture Testing, Governance & Audit	Testing and control

Architecture Statement

We recommend putting the following almost verbatim into the technical SOW:

TICVAI shall implement a media-independent Virtual Ticket architecture. Every issued ticket shall have one persistent Virtual Ticket ID representing the authoritative ticket, entitlement, validity and usage record. A Virtual Ticket may simultaneously bind to zero, one or multiple credential media, including QR/barcode, dynamic QR, RFID, NFC, Apple Wallet, Google Wallet, printed media, cards, wristbands, biometric/face-recognition references and future credential technologies.

Credential media shall not create independent copies of the ticket entitlement. All media associated with the same Virtual Ticket shall resolve to the same authoritative ticket and entitlement state through a centralized credential-resolution mechanism. Replacing, revoking, regenerating or rebinding a credential medium shall not, by itself, require replacement of the underlying Virtual Ticket.

Board 1 configuration flow

Virtual Ticket Identity → Lifecycle → Media Registry → Multi-Media Binding → Credential Resolution → Entitlement Synchronization → Activation/Fallback → Replacement/Rebinding → Test & Govern

This gives us the correct foundation. **Board 2 can now concentrate completely on the Multi-Format Ticket Media Design Studio**—including dedicated design/configuration experiences for QR tickets, PDF/print, Apple Wallet, Google Wallet, RFID/NFC cards and

wristbands, digital cards and other media—without mixing those design functions into the core Virtual Ticket architecture.

Area 15 — Ticket Media & Credential Management

Board 2 — Ticket Media Design Studio & Template Management

10 Backend Screens — Detailed Scope of Work

The screenshot displays the TICVAI Board 2 interface, titled "AREA 15 - TICKET MEDIA & CREDENTIAL MANAGEMENT | BOARD 2 - TICKET MEDIA DESIGN STUDIO & TEMPLATE MANAGEMENT (10 SCREENS)". The interface is divided into a left sidebar with navigation options and a main content area with 10 screens.

Left Sidebar:

- Dashboard**
- REVENUE**
 - Commercial Pricing
 - Promotions
 - Tax, Fees & Calculation
- PRODUCTS & CONTENT**
 - Products
 - Events & Performances
 - Experiences
 - Seat / Resource Mgmt
- ORDERS & TICKETING**
 - Orders
 - Ticketing
 - Fulfillment
- 15 Ticket Media & Credential**
 - 15.1 Architecture
 - 15.2 Design Studio (Board 2)
 - 15.3 Operations (Board 3)
- ACCESS & VALIDATION**
 - Access Control
 - Validation Devices
 - On-Site Operations
- CUSTOMERS & MEMBERSHIPS**
 - Customers
 - Memberships
 - Loyalty
- PLATFORM & CONFIG**
 - Channels
 - Brands & Venues
 - Users & Roles
 - System Settings
- REPORTS & ANALYTICS**
 - Audit Logs
 - Support Tools

Main Content Area Screens:

- 15.2.1 Media Design Studio Command Center:** Overview of templates with metrics: Total Templates (286), Published (178), Draft (64), Pending Approval (18). Includes a "Recent Templates" table.
- 15.2.2 Digital QR & Barcode Ticket Designer:** Design interface for a "VIP Concert Mobile v4 (QR)" ticket, showing fields for event name, date, time, and QR code.
- 15.2.3 PDF, Printable & POS Ticket Designer:** Design interface for a "Waterpark A4 Ticket v5 (PDF)" ticket, showing fields for event name, date, time, and barcode.
- 15.2.4 Apple Wallet Pass Designer:** Design interface for an "Apple Wallet Concert v2" pass, showing fields for event name, date, time, and QR code.
- 15.2.5 Google Wallet Pass Designer:** Design interface for a "Google Wallet Concert v1" pass, showing fields for event name, date, time, and QR code.
- 15.2.6 RFID, NFC, Card & Wristband Media Designer:** Design interface for a "Gold Annual Pass RFID v3" pass, showing fields for event name, date, time, and QR code.
- 15.2.7 Digital Card, Membership & Wearable Designer:** Design interface for a "Platinum Membership Card v2" card, showing fields for member name, ID, and QR code.
- 15.2.8 Dynamic Fields, Data Mapping & Content Builder:** Configuration interface for dynamic fields and data mapping.
- 15.2.9 Branding, Localization & Template Inheritance:** Configuration interface for branding and localization.
- 15.2.10 Multi-Media Preview, Testing, Approval & Publication:** Overview of the approval and publication workflow.

Board Objective

Board 1 established **what the Virtual Ticket is** and how multiple credentials can point to the same authoritative ticket.

Board 2 defines **how each media type is created, designed, configured, branded, populated with data, previewed, tested and published.**

The configuration flow is:

Select Media Type → Select/Create Template → Design Layout → Add Dynamic Data → Configure Media-Specific Properties → Brand & Localize → Associate Usage → Preview → Validate → Approve & Publish

A key principle:

TICVAI must not use one generic ticket designer for every credential technology.

TICVAI should provide a common **Media Design Studio framework**, but the configuration tools presented to the administrator must adapt to the selected media type.

For example, an Apple Wallet pass requires different configuration from an RFID wristband, while both can represent the same Virtual Ticket.

15.2.1 — Media Design Studio Command Center

Purpose

Provide administrators with the central workspace for creating and managing all ticket and credential media templates.

This should be the entry point for the entire no-code Media Design Studio.

Executive KPIs

Display:

- Total Media Templates
- Published
- Draft
- Pending Approval
- Scheduled
- Archived
- QR / Digital Templates
- PDF / Print Templates
- Apple Wallet Templates
- Google Wallet Templates
- RFID / NFC Templates
- Card / Wristband Templates
- Templates Requiring Review
- Templates with Validation Errors

Template Directory

Each template should show:

- Template ID
- Template Name
- Media Type
- Brand
- Venue
- Product / Event association
- Language
- Version
- Effective From
- Effective To
- Status
- Owner
- Last Modified

Create Media

Primary action:

+ Create Media Template

The administrator first selects the media:

Digital

- QR Ticket
- Dynamic QR Ticket
- Barcode Ticket
- Mobile Ticket

Print

- PDF
- A4 / A5
- Thermal
- POS
- Custom Print

Wallet

- Apple Wallet
- Google Wallet

Physical

- RFID Card
- RFID Wristband
- NFC Card
- NFC Wristband
- Membership Card
- Custom Wearable

Template Sources

Allow:

- Blank Template
- Duplicate Existing
- Use Corporate Template
- Use Venue Template
- Use Product Template
- Import supported template definition

AI Capability

AI can recommend existing templates rather than creating unnecessary duplicates.

Example:

“Dubai VIP Mobile Ticket v4 already matches 92% of this configuration.”

Acceptance Condition

Administrators can locate, create and manage every ticket-media template from one centralized Media Design Studio.

15.2.2 — Digital QR & Barcode Ticket Designer

Purpose

Provide a visual no-code designer specifically for digital QR and barcode tickets.

Designer Canvas

Drag-and-drop components such as:

- Logo
- Event Image
- Event Name
- Ticket Type
- Customer Name
- Participant Name
- Date
- Time
- Venue
- Entrance
- Section
- Row
- Seat
- Price
- Order Reference
- Virtual Ticket ID
- QR
- Barcode
- Terms
- Instructions
- Waiver Link / Status
- Sponsor
- Custom Fields

QR Configuration**Configure:**

- Static QR
- Dynamic QR
- Signed QR
- Tokenized QR
- Rotation behavior where supported
- Size
- Position
- Quiet zone
- Error correction
- Expiration
- Refresh behavior

The designer configures presentation and payload profile; **Board 1 remains responsible for resolving the credential to the Virtual Ticket.**

Barcode Configuration

Support configured symbologies as required by the solution/integration.

Configure:

- Barcode type
- Orientation
- Size
- Human-readable value
- Hide/show encoded reference

Price Visibility**Important existing TICVAI requirement:****Show Price: ON / OFF**

This can depend on template or delivery context.

Responsive Preview**Preview:**

Mobile | Tablet | Desktop

Acceptance Condition

Authorized administrators can visually design and configure customer-facing QR/barcode tickets without development.

15.2.3 — PDF, Printable & POS Ticket Designer

Purpose

Design tickets intended for printing, PDF generation, POS, box office and other physical/document outputs.

Document Formats**Support:**

- PDF
- A4
- A5
- Custom dimensions
- POS receipt
- Thermal ticket
- Box-office stock
- Pre-printed stock where required

Visual Designer**Configure:**

- Page size
- Orientation
- Margins
- Header
- Footer
- Background
- Logo
- Images

- Text
- Dynamic fields
- QR/barcode
- Terms
- Perforation indicators where applicable
- Print-safe zones

Front / Back

Where supported:

Front Designer

and

Back Designer

Printer Considerations

Configure/reference:

- Printer profile
- DPI
- Paper/stock type
- Thermal layout
- Cut behavior
- Print margins
- Supported printer integration

Virtual On-Screen Print

Support the requirement for an operator to preview the **actual generated ticket on screen** without physically printing it.

Sample Data

Administrators can populate the template with test data before publication.

Acceptance Condition

TICVAI can create governed print/PDF/POS ticket layouts for different devices, paper sizes and operational environments.

15.2.4 — Apple Wallet Pass Designer

Purpose

Provide an Apple Wallet-specific configuration experience for eligible TICVAI credentials.

This should **not simply be a PDF ticket rendered inside a wallet**.

Pass Configuration

Configure the applicable wallet-pass structure, such as:

- Pass identity
- Organization
- Description
- Logo
- Icon
- Images where supported

- Background appearance
- Foreground appearance
- Label appearance
- Primary fields
- Secondary fields
- Auxiliary fields
- Header fields
- Back fields

Dynamic Data

Map fields such as:

Event Name
 Date / Time
 Venue
 Ticket Holder
 Ticket Type
 Seat
 Entrance
 Membership
 Virtual Ticket Reference

Credential

Configure the appropriate:

- QR
- Barcode
- Token/reference

according to the credential profile.

Pass Updates

Configure/reference:

- Dynamic updates
- Event changes
- Seat changes
- Ticket status changes
- Relevant notification/update behavior
- Expiry
- Revocation/invalidation behavior where supported

Preview

Provide an appropriate wallet-style preview before publication.

Important Architecture

The Apple Wallet pass is a **media representation**.

It must resolve to:

Virtual Ticket ID

It does not become an independent ticket.

Acceptance Condition

Administrators can configure, preview, test and publish Apple Wallet presentations of TICVAI Virtual Tickets using supported wallet capabilities.

15.2.5 — Google Wallet Pass Designer

Purpose

Provide a dedicated Google Wallet configuration environment.

Wallet Configuration

Configure supported:

- Pass class/template
- Issuer
- Title
- Logo
- Hero/image assets where applicable
- Event information
- Venue
- Date/time
- Ticket holder
- Seat
- Ticket type
- Custom fields
- Status
- Links
- Additional information

Credential Presentation

Configure:

- QR
- Barcode
- Credential/token reference

according to supported wallet functionality.

Dynamic Updates

Support/reference updates such as:

- Event time change
- Venue change
- Seat reassignment
- Ticket status change
- Relevant ticket information changes

Preview

Provide device-oriented preview before publication.

Template Reuse

Allow one Google Wallet template to support multiple products through dynamic data mapping.

Acceptance Condition

Administrators can independently configure and govern Google Wallet ticket presentations while retaining the same underlying Virtual Ticket architecture.

15.2.6 — RFID, NFC, Card & Wristband Media Designer

Purpose

Configure both the **visual and technical profile** of physical electronic credentials.

This is important because RFID/NFC media are not merely artwork.

Media Types

Support:

- RFID Card
- RFID Wristband
- NFC Card
- NFC Wristband
- Membership Card
- Staff/guest card where applicable
- Custom Wearable

Physical Media Configuration

Configure:

- Media dimensions
- Front
- Back
- Printable area
- Logo
- Customer name
- Photo
- Membership tier
- Expiry
- Serial number
- QR/barcode if combined
- Custom artwork
- Sponsor/venue branding

Technology Profile

Configure/reference:

- RFID/NFC technology
- Chip/profile
- UID/reference handling
- Encoding profile
- Provider
- Reader compatibility

- Printer/encoder integration

Encoding

Define which secure reference/token is encoded or associated.

The media should resolve to:

Credential Binding → Virtual Ticket

rather than storing a complete independent ticket entitlement on the media.

Wristband Configuration

Support:

- Disposable
- Reusable
- Printed
- Encoded
- Color/category
- Size where applicable
- Activation at collection
- Deposit/reference where applicable

Acceptance Condition

TICVAI can configure the physical appearance and technical credential profile of RFID/NFC cards, wristbands and supported wearable media.

15.2.7 — Digital Card, Membership & Wearable Designer

Purpose

Provide specialized configuration for persistent credentials that may represent longer-lived relationships rather than a single event ticket.

Examples include:

- Annual Pass
- Membership Card
- Season Pass
- VIP Credential
- Staff/Partner Credential where applicable
- Digital Membership Card
- Custom Wearable

Digital Card Designer

Configure:

- Brand
- Card artwork
- Member photograph
- Member name
- Membership number
- Tier
- Validity

- QR/barcode
- Status
- Benefits summary
- Dynamic messaging

Physical Card Variants

Support different:

- Tier designs
- Membership types
- Brands
- Venues
- Age categories

Example:

Silver → Silver artwork

Gold → Gold artwork

Platinum → Platinum artwork

Wearable Templates

Configure visual and technical characteristics for supported wearables.

Dynamic State

Persistent media may change presentation based on:

- Membership renewal
- Tier change
- Expiry
- Suspension
- Benefit status

Acceptance Condition

TICVAI can design and maintain reusable digital-card, membership-card and wearable templates linked to persistent Virtual Tickets or applicable entitlement identities.

15.2.8 — Dynamic Fields, Data Mapping & Content Builder

Purpose

Create a centralized reusable field library so development team does **not hard-code ticket fields separately into every media designer**.

This is another important architecture screen.

Field Categories

Customer

- Customer Name
- Customer ID
- Mobile
- Email
- Photo

Participant

- Participant Name
- DOB/Age category
- Participant ID

Order

- Order ID
- Booking Reference
- Purchase Date
- Channel

Virtual Ticket

- Virtual Ticket ID
- Ticket Type
- Status
- Validity
- Usage Status

Product/Event

- Product
- Event
- Performance
- Date
- Time
- Venue
- Entrance

Seating

- Section
- Block
- Row
- Seat

Commercial

- Face Value
- Paid Price
- Discount
- Currency

Membership

- Membership ID
- Tier
- Expiry

Compliance

- Waiver Status
- Waiver Link where permitted

Credential

- QR

- Barcode
- Credential Reference

Custom Fields

Authorized tenants should be able to define additional approved fields.

Data Mapping

Example:

{{ticket.seat}} → A18

{{event.start_time}} → 19:30

{{customer.name}} → Ahmed Hassan

Formatting

Configure:

- Date formats
- Time formats
- Currency
- Number formatting
- Text transformation
- Character limit
- Conditional visibility

Conditional Content

Example:

IF:

Seat Assigned = Yes

show:

SECTION / ROW / SEAT

Otherwise hide the section.

Sensitive Data Control

Not every field should be available on every media.

For example, sensitive personal information should not automatically be permitted on a printed ticket.

AI Assistance

AI may recommend useful fields and detect potential mapping problems.

Acceptance Condition

All media designers consume a governed reusable field/data-mapping framework rather than independently hard-coding ticket information.

15.2.9 — Branding, Localization & Template Inheritance

Purpose

Allow TICVAI's multi-tenant clients to control branding and localization without rebuilding every ticket template.

Brand Hierarchy

Support inheritance such as:

Tenant / Corporate

↓

Brand

↓

Venue

↓

Event

↓

Product

↓

Media Template**Inheritance Example**

TICVAI Corporate Template

↓

Dubai Parks Brand

↓

Waterpark

↓

VIP Admission

↓

Apple Wallet

A lower level can inherit approved elements while overriding permitted properties.

Brand Components**Configure:**

- Logo
- Colors
- Typography
- Backgrounds
- Header/footer
- Legal footer
- Support information
- Sponsor placement
- Images

Localization**Support:**

- English
- Arabic
- Additional configured languages

RTL**Arabic templates must properly support:**

Right-to-Left layout, not simply translated text.

Translation Management

Track:

- Source language
- Translation
- Translation status
- Reviewer
- Approval
- Version

Shared vs Media-Specific Branding

For example:

Corporate logo can be inherited across:

QR
PDF
Apple Wallet
Google Wallet
RFID

while each medium retains its technology-specific layout.

Acceptance Condition

TICVAI can centrally manage multi-brand and multilingual ticket-media presentation with controlled inheritance and media-specific overrides.

15.2.10 — Multi-Media Preview, Testing, Approval & Publication

Purpose

Provide the final quality and governance gate before any media template becomes operational.

This should be a particularly visual screen.

Universal Preview

Administrators select:

Virtual Ticket Sample

and TICVAI renders the available media.

For example:

VIP Concert — Sample Ticket

Then show:

Mobile QR

PDF

Apple Wallet

Google Wallet

RFID Card

Wristband

side by side where practical.

Sample Customer

Example:**Ahmed Hassan**

Virtual Ticket: VT-2026-009821

Product: VIP Concert

Date: 15 Sep 2026

Time: 19:30

Venue: Arena

Section: A

Row: 3

Seat: 18

Testing**Validate:**

- Dynamic fields
- Missing fields
- QR readability
- Barcode readability
- Wallet configuration
- Print boundaries
- Localization
- RTL
- Branding
- Image resolution
- Credential payload
- Virtual Ticket resolution
- Provider configuration

Device Testing**Where applicable:**

- Desktop
- Mobile
- Tablet
- Printer
- POS
- Wallet
- Encoder

Pre-Publication Validation**TICVAI should automatically detect:**

Missing mandatory field.

QR overlaps customer name.

Arabic layout exceeds printable area.

Wallet credential has no valid payload mapping.

RFID template has no encoder profile.

Approval Workflow**Recommended:**

Designer

- Brand Review
- Operations Review
- Technical/Security Review where applicable
- Approval
- Publication

Publication Options**Support:**

- Publish Now
- Schedule
- Selected Brands
- Selected Venues
- Selected Products
- Selected Channels
- Controlled rollout

Versioning

Publishing a new template should create a governed version.

Existing issued credentials should not automatically be historically rewritten simply because a template changes.

AI Pre-Publication Review

AI can identify usability/configuration concerns.

Example:

“The ticket holder name may be truncated on smaller mobile screens in Arabic.”

or:

“The QR occupies only 9% of the printable area and may be difficult to scan on the selected thermal printer profile.”

Human approval remains required according to configured governance.

Acceptance Condition

Administrators can preview, test, approve, version and publish every supported ticket-media configuration before it becomes available for production issuance.

Board 2 — Final Screen Register

Screen	Backend Screen	Primary Responsibility
15.2.1	Media Design Studio Command Center	Template portfolio
15.2.2	Digital QR & Barcode Ticket Designer	Digital ticket design
15.2.	PDF, Printable & POS Ticket Designer	Print/PDF media

Screen	Backend Screen	Primary Responsibility
3		
15.2.4	Apple Wallet Pass Designer	Apple Wallet
15.2.5	Google Wallet Pass Designer	Google Wallet
15.2.6	RFID, NFC, Card & Wristband Media Designer	Physical/electronic media
15.2.7	Digital Card, Membership & Wearable Designer	Persistent credentials
15.2.8	Dynamic Fields, Data Mapping & Content Builder	Reusable data layer
15.2.9	Branding, Localization & Template Inheritance	Brand/language governance
15.2.10	Multi-Media Preview, Testing, Approval & Publication	QA and publication

Critical Requirements

There are five requirements I would explicitly freeze for this board:

- 1. No-code media design.** Authorized TICVAI administrators must be able to create and maintain ticket-media templates without software development.
- 2. Media-specific designers.** QR/PDF, Apple Wallet, Google Wallet, RFID/NFC and wearables must expose configuration appropriate to their technology rather than being forced through one generic layout editor.
- 3. One Virtual Ticket behind every medium.** The designer controls the **presentation and media configuration**; it does not create a separate ticket entitlement.
- 4. Central dynamic-field framework.** Ticket/customer/event/seat/order/membership fields must come from governed reusable data mappings rather than being hard-coded into each template.
- 5. Extensible media framework.** Future credential technologies must be addable without redesigning the Virtual Ticket core.

Board 2 Architecture

Virtual Ticket → Media Template Set → Media-Specific Designer → Dynamic Data Mapping → Branding/Localization → Technology Configuration → Preview/Test → Approval → Published Media Template

For example, one **VIP Concert Virtual Ticket** could use a template set containing:

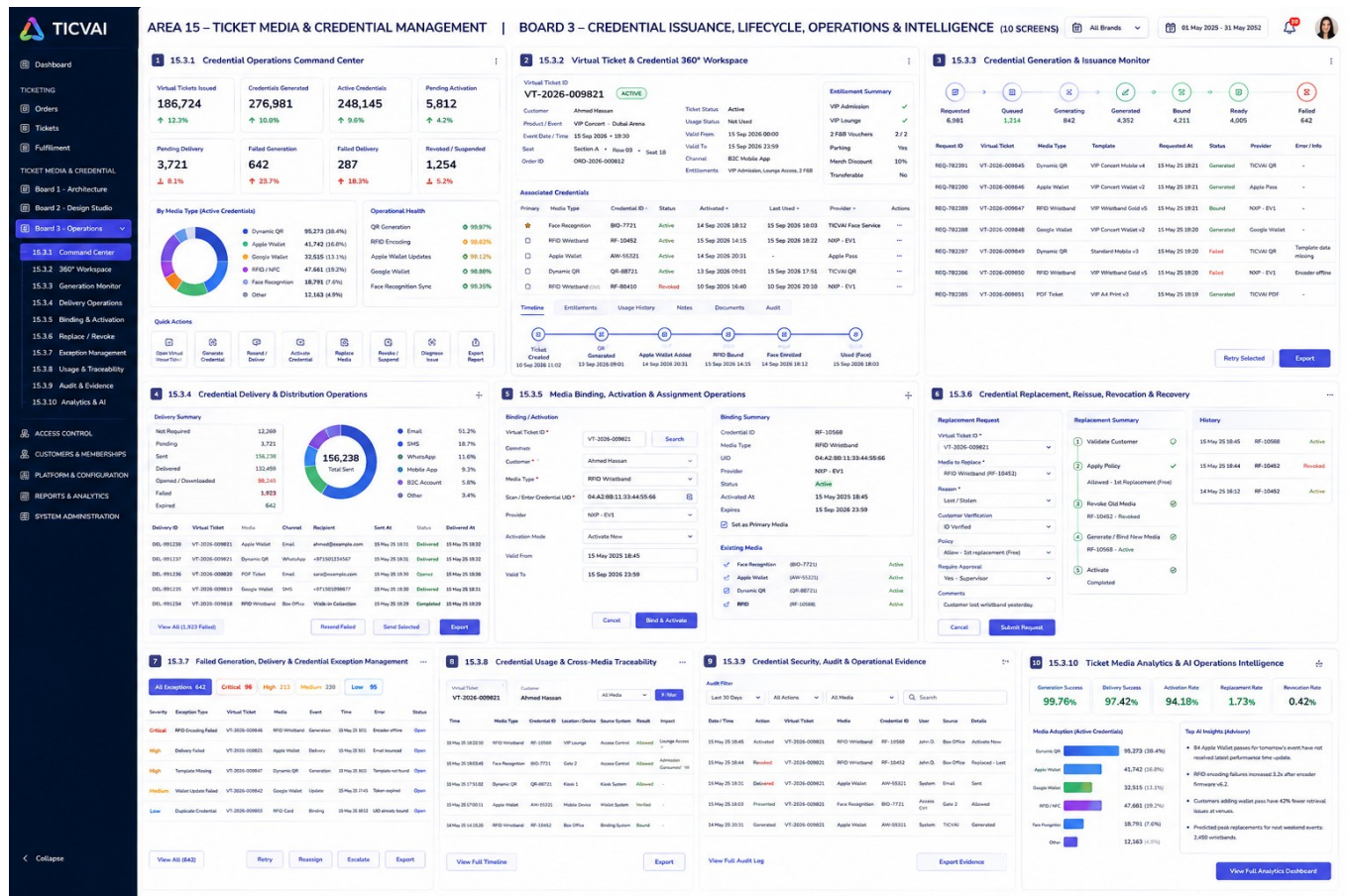
Mobile Dynamic QR + PDF + Apple Wallet + Google Wallet + VIP RFID Wristband

All five may look and behave differently, but **all five represent the same Virtual Ticket**.

This completes the **creation/design layer**. Board 3 will then handle the real operational lifecycle: **generate → issue → deliver → bind → activate → replace/revoke → troubleshoot → trace → audit → analyze**.

Area 15 — Ticket Media & Credential Management

Board 3 — Credential Issuance, Lifecycle, Operations & Intelligence



10 Backend Screens — Detailed Scope of Work

Board Objective

Board 1 established the **Virtual Ticket as the authoritative ticket identity** and the one-Virtual-Ticket-to-many-media architecture.

Board 2 established how administrators **design and configure each media type**.

Board 3 manages what happens to the **actual credential instances in production** after a Virtual Ticket has been created.

The operational lifecycle is:

Virtual Ticket Created → Determine Required Media → Generate Credential → Bind to Virtual Ticket → Deliver / Assign → Activate → Monitor → Use → Replace / Revoke if Required → Expire → Audit

The critical architecture remains:

Virtual Ticket = authoritative ticket and entitlement
Credential Media = representations of that Virtual Ticket

Therefore, Board 3 must never create independent entitlement states for QR, RFID, Wallet, Face, etc.

15.3.1 — Credential Operations Command Center

Purpose

Provide Operations, Ticketing, Customer Service and Technical teams with a real-time command center covering all issued credential media.

This is the operational starting point for Area 15.

Executive KPIs

Display:

- Virtual Tickets Issued
- Credentials Generated
- Active Credentials
- Pending Generation
- Pending Delivery
- Pending Binding
- Pending Activation
- Suspended
- Revoked
- Expired
- Failed Generation
- Failed Delivery
- Synchronization Exceptions
- Multi-Media Virtual Tickets
- Virtual Tickets Without Active Media

Media Breakdown

Display credentials by:

- Dynamic QR
- Barcode
- PDF
- Apple Wallet
- Google Wallet
- RFID
- NFC
- Face Recognition Reference
- Card
- Wristband
- Other

Operational Queue

Each record should show:

- Virtual Ticket ID
- Credential ID
- Media Type
- Customer / Participant

- Product
- Event
- Credential Status
- Delivery Status
- Activation Status
- Binding Status
- Provider
- Last Activity
- Exception
- Owner

Operational Health

Provide indicators such as:

QR Generation — 99.98% Healthy

RFID Encoding — 98.7% Healthy

Apple Wallet Updates — Attention

Google Wallet — Healthy

Filters

Filter by:

- Brand
- Venue
- Event
- Product
- Date
- Media
- Provider
- Status
- Channel
- Exception
- Customer

Quick Actions

Authorized users can:

- Open Virtual Ticket
- Generate Credential
- Resend
- Activate
- Suspend Media
- Replace
- Revoke
- Diagnose
- View History

AI Capability

AI should prioritize operational problems.

Example:

“84 Apple Wallet credentials for tomorrow's event have not received the latest performance-time update.”

Acceptance Condition

Operations can understand the health, status and exceptions of all issued credentials from one central workspace.

15.3.2 — Virtual Ticket & Credential 360° Workspace

Purpose

Provide a complete operational view of one Virtual Ticket and **every media credential currently or historically associated with it.**

This is one of the most important operational screens.

Virtual Ticket Header

Display:

- Virtual Ticket ID
- Ticket Holder
- Participant
- Product
- Event
- Performance
- Venue
- Seat
- Order
- Ticket Status
- Usage Status
- Validity
- Entitlements

Credential Wallet

Show all associated credentials.

Example:

VT-2026-009821 — ACTIVE

Media	Credenti al	Status
Dynamic QR	QR-88721	Active
RFID Wristband	RF-10452	Active
Apple Wallet	AW-55321	Active

Media	Credenti al	Status
Face	BIO-7721	Active
Old RFID	RF-88410	Revoke d

Primary / Secondary Media

Identify:

- Primary
- Secondary
- Fallback
- Temporary
- Revoked historical media

Credential Details

For each media show:

- Binding ID
- Media type
- Provider
- Issued
- Delivered
- Activated
- Valid From/To
- Last update
- Last presentation/use
- Device/reference where appropriate
- Status

Operational Actions

Depending on permission:

- Generate Additional Media
- Bind RFID
- Add Wallet Pass
- Initiate Face Enrollment
- Replace Media
- Suspend
- Revoke
- Resend
- Refresh
- Diagnose

Timeline

Provide one chronological timeline:

Ticket Created → QR Generated → Apple Wallet Added → RFID Assigned → Face Enrolled → Entry via Face → RFID Replaced...

Acceptance Condition

Staff can view and manage the complete multi-media credential history of a Virtual Ticket without treating each credential as a separate ticket.

15.3.3 — Credential Generation & Issuance Monitor

Purpose

Manage and monitor generation of credential instances from approved Board 2 media templates.

Generation Sources

Credential generation may be triggered by:

- Order Confirmation
- Ticket Issuance
- Membership Activation
- Customer Request
- Staff Action
- RFID Collection
- Wallet Request
- Face Enrollment
- API
- Bulk Operation
- Scheduled Process

Generation Pipeline

Show:

Requested → Queued → Template Resolved → Data Mapped → Credential Generated → Bound → Ready → Delivered

Generation Record

Display:

- Request ID
- Virtual Ticket
- Media
- Template
- Template Version
- Product
- Customer
- Trigger
- Provider
- Requested At
- Generated At
- Status

- Error

Template Resolution

TICVAI determines the correct template based on:

- Brand
- Venue
- Product
- Event
- Media
- Channel
- Language
- Customer context

Bulk Generation

Support controlled bulk operations.

Examples:

- 5,000 event QR tickets
- 2,000 school-camp wristbands
- 750 membership cards

Failure Handling

Failures may include:

- Template missing
- Required data missing
- Provider unavailable
- Invalid payload
- Token generation failure
- Wallet generation failure
- Encoder unavailable

Retry

Support:

- Automatic Retry
- Manual Retry
- Retry Selected
- Retry All Eligible
- Escalate

AI Capability

AI can identify abnormal failure patterns.

“RFID issuance failure increased from 0.8% to 7.2% after Encoder Profile v4.1 was activated.”

Acceptance Condition

TICVAI can reliably generate credential instances from approved templates and provide full operational visibility into successful and failed issuance.

15.3.4 — Credential Delivery & Distribution Operations

Purpose

Manage how generated ticket media are delivered or made available to customers, participants and operational staff.

Delivery Channels

Support as applicable:

- Email
- SMS Link
- WhatsApp integration
- B2C Account
- Mobile App
- Download
- Apple Wallet
- Google Wallet
- POS
- Box Office
- Kiosk
- Group Portal
- API
- Physical Collection

Delivery Status

Recommended lifecycle:

Not Required → Pending → Sent → Delivered → Opened / Downloaded → Completed

Alternative states:

- Failed
- Bounced
- Expired
- Cancelled

Distribution Record

Show:

- Virtual Ticket
- Media
- Recipient
- Channel
- Destination
- Sent At
- Delivered At
- Opened/Downloaded
- Attempt
- Status

Multi-Recipient Rules

Where allowed, support:

- Purchaser
- Ticket Holder
- Participant
- Guardian
- Group Leader
- Authorized recipient

Bulk Distribution**Support:**

- Send All
- Send Selected
- Resend Pending
- Resend Failed
- Send Group
- Export Status

Security**Delivery must respect configured:**

- Authentication
- Link expiration
- Single/multiple use
- Customer identity
- Token security
- Data masking

Acceptance Condition

TICVAI can securely distribute credentials through supported channels and provide complete delivery tracking.

15.3.5 — Media Binding, Activation & Assignment Operations

Purpose

Manage credentials that require operational assignment or activation after ticket issuance.

This is especially important for:

- RFID
- NFC
- Wristbands
- Physical cards
- Face recognition
- Temporary credentials

Assignment Workflow**Example:**

Customer arrives.

Scan Virtual Ticket QR**Select RFID Wristband****Scan Wristband UID****Resolve Virtual Ticket****Bind****Activate****Ready****Binding Workspace****Display:**

- Virtual Ticket
- Customer
- Product
- Existing Media
- New Media Type
- Credential ID / UID
- Provider
- Activation
- Validity
- Binding Rule

RFID/NFC Assignment**Support:**

- Scan
- Tap
- Manual lookup where authorized
- Batch assignment
- Encoder assignment

Face Enrollment**Where integrated:****Virtual Ticket → Participant → Biometric Provider Reference → Binding**

TICVAI should retain the authorized reference/binding rather than unnecessarily duplicating raw biometric data.

Activation Controls**Allow:**

- Activate Now
- Schedule

- Activate on First Use
- Activate on Collection
- Temporary Activation

Duplicate Detection

Before binding:

“This RFID credential is already bound to VT-008882.”

Block or require authorized resolution.

Acceptance Condition

Operational staff can securely assign and activate supported physical, digital and biometric credential media against the correct Virtual Ticket.

15.3.6 — Credential Replacement, Reissue, Revocation & Recovery

Purpose

Manage operational credential changes while preserving the underlying Virtual Ticket.

Scenarios

Support:

- Lost
- Stolen
- Damaged
- Compromised
- Customer changed phone
- RFID failure
- Wristband replacement
- QR compromise
- Wallet replacement
- Face re-enrollment
- Incorrect assignment

Replacement Flow

Recommended:

Locate Virtual Ticket

- Select Credential
- Replacement Reason
- Verify Customer
- Apply Policy
- Approval if required
- Revoke/Suspend Old Media
- Generate/Bind New Media
- Activate
- Confirm

Example**Virtual Ticket:****VT-009821 — ACTIVE****Old RFID:****RF-88721 — REVOKED****New RFID:****RF-99211 — ACTIVE****QR:****QR-55128 — ACTIVE**

The Virtual Ticket remains unchanged.

Replacement Fees

Where applicable, reference pricing/fee configuration rather than hard-code financial calculations here.

Security Rules**Configure/reference:**

- Immediate old-media revocation
- Grace period
- Maximum replacements
- Identity verification
- Supervisor approval
- Reason codes

Recovery

Allow a suspended credential to be restored only where policy permits.

Acceptance Condition

TICVAI can replace or recover credential media without unnecessarily replacing the Virtual Ticket and without losing historical traceability.

15.3.7 — Failed Generation, Delivery & Credential Exception Management

Purpose

Provide one dedicated operational queue for credential-related failures.

Exception Categories**Include:**

- Generation Failed
- Binding Failed
- Activation Failed
- Delivery Failed
- Wallet Failure
- RFID Encoding Failure
- Duplicate Credential
- Invalid Token

- Provider Failure
- Synchronization Failure
- Missing Template
- Missing Required Data
- Expired Credential
- Mapping Failure
- Unknown Credential

Exception Queue

Display:

- Severity
- Virtual Ticket
- Credential
- Media
- Customer
- Event
- Failure
- Time
- Operational Impact
- Retry Status
- Owner

Prioritization

Consider:

- Event proximity
- Customer arrival time
- Number of affected tickets
- No alternative media
- VIP/customer-service impact
- Access impact
- Provider outage

Corrective Actions

Allow:

- Retry
- Regenerate
- Resend
- Rebind
- Switch Media
- Use Fallback
- Escalate
- Assign Owner
- Open Technical Case

AI Root-Cause Assistance

AI may correlate failures.

Example:

“91% of today's RFID encoding failures originate from Encoder Group 3 after firmware update 6.2.”

Acceptance Condition

Credential failures are centrally identified, prioritized and resolved before unnecessarily impacting customer admission or fulfillment.

15.3.8 — Credential Usage & Cross-Media Traceability

Purpose

Provide end-to-end visibility into how the different media attached to one Virtual Ticket have been presented or used.

This screen is for **credential traceability**, while Area 16 remains responsible for the actual access-control decision.

Usage Timeline

Example:

VT-009821

09:02 — Apple Wallet added
 14:15 — RFID bound
 17:51 — Mobile QR presented at Kiosk
 18:03 — Face credential presented at Gate 2
 18:03 — Access result received: Allowed
 18:04 — Virtual Ticket usage updated
 18:22 — RFID presented at VIP Lounge

Usage Record

Capture/reference:

- Virtual Ticket
- Credential
- Media
- Presentation timestamp
- Location
- Device
- External system
- Transaction type
- Result
- Entitlement impact
- Synchronization status

Cross-Media View

Allow administrators to understand:

Which media was used?

When?

Where?

Which Virtual Ticket did it resolve to?

What happened to the master entitlement?

Important Boundary

Area 15 records and displays the credential event.

Area 16 decides admission.

For example:

Area 15:

RFID RF-10028 → VT-009821

Area 16:

VT-009821 → Gate 4 → ALLOW

Area 15 then retains the linked operational trace.

Acceptance Condition

TICVAI provides complete cross-media traceability while preserving Access Control as the authoritative admission decision engine.

15.3.9 — Credential Security, Audit & Operational Evidence

Purpose

Maintain complete evidence of credential creation and lifecycle activity.

Audit Events

Capture:

- Credential Requested
- Generated
- Bound
- Delivered
- Activated
- Updated
- Presented
- Suspended
- Reactivated
- Replaced
- Revoked
- Expired
- Rebound
- Regenerated
- Deleted where legally/permissibly applicable

Audit Record

Record:

- Virtual Ticket
- Credential
- Media
- Action

- Before
- After
- Actor
- Source
- Device
- Date/time
- Reason
- Approval
- Provider reference
- Related transaction

Credential Chain

Example:

VT-009821

RFID-1001

↓ Lost

RFID-1001 — Revoked

↓ Replacement

RFID-1057 — Active

This entire history must remain reconstructable.

Security Monitoring

Identify:

- Excessive regeneration
- Repeated replacement
- Suspicious rebinding
- Multiple credential assignments
- Unexpected provider/token changes
- Unauthorized administrative actions

Access Control

Audit information must be governed according to:

- Role
- Tenant
- Venue
- Business Unit
- Data sensitivity

Evidence Export

Authorized users may generate credential-history evidence for:

- Customer dispute
- Operational investigation
- Fraud investigation
- Audit
- Compliance

Acceptance Condition

Every significant credential lifecycle action is traceable with sufficient evidence to reconstruct who did what, when, why and to which Virtual Ticket.

15.3.10 — Ticket Media Analytics & AI Operations Intelligence**Purpose**

Provide management and operations with analytics and AI intelligence across Virtual Tickets and credential media.

This should be a serious operational intelligence layer—not simply a chatbot.

Executive KPIs**Analyze:**

- Credentials Generated
- Generation Success %
- Delivery Success %
- Activation %
- Wallet Adoption
- RFID Adoption
- Face Credential Adoption
- Multi-Media Adoption
- Replacement Rate
- Revocation Rate
- Generation Failure %
- Delivery Failure %
- Average Generation Time
- Average Resolution Time
- Media Usage Distribution

Media Adoption**Example:****Eligible customers**

Mobile QR — 78%

Apple Wallet — 31%

Google Wallet — 22%

RFID — 18%

Face — 12%

Multiple media may apply to one Virtual Ticket, so percentages do not need to total 100%.

Operational Analytics**Analyze by:**

- Brand
- Venue
- Event
- Product
- Channel

- Media
- Provider
- Device
- Customer segment
- Time period

Reliability Analytics

Compare:

- Provider uptime
- Generation failures
- Encoding failures
- Delivery failures
- Synchronization delay
- Replacement frequency

AI Operational Intelligence

AI can identify patterns such as:

“RFID wristbands issued by Encoder Group B have a 3.8× higher replacement rate than other encoder groups.”

or:

“Customers who add their ticket to a mobile wallet have 42% fewer ticket-retrieval issues at the venue.”

Predictive Operations

Where sufficient data exists, models may predict:

- Credential failure risk
- Delivery failure probability
- Media demand for upcoming events
- RFID/wristband stock requirements
- Operational workload
- Likely on-site replacement volumes

Natural-Language Analytics

Authorized users could ask:

“Which credential type had the most failures yesterday?”

“Show tomorrow's events with customers who have no active usable media.”

“Which RFID provider has the highest replacement rate?”

“How many annual-pass holders use both face and RFID?”

AI Recommendations

Examples:

“Pre-generate QR credentials for 4,200 attendees before the forecast peak issuance window.”

“Investigate RFID Encoder Group 3.”

“Send wallet-add reminders to customers for tomorrow's event.”

AI Governance

AI may:

- Detect
- Explain
- Prioritize
- Predict
- Recommend
- Draft operational actions

AI must not independently:

- Create entitlement
- Change ticket ownership
- Fabricate credential usage
- Mark unused tickets as consumed
- Bypass credential security
- Override access-control decisions
- Rebind sensitive credentials without governed authorization

Acceptance Condition

TICVAI provides explainable analytics and AI intelligence that improves credential reliability and operational performance without transferring authoritative ticket or security decisions to generative AI.

Board 3 — Final Screen Register

Screen	Backend Screen	Primary Responsibility
15.3.1	Credential Operations Command Center	Overall operational health
15.3.2	Virtual Ticket & Credential 360° Workspace	Complete ticket/media view
15.3.3	Credential Generation & Issuance Monitor	Generate credentials
15.3.4	Credential Delivery & Distribution Operations	Deliver credentials
15.3.5	Media Binding, Activation & Assignment Operations	Assign/activate media
15.3.6	Credential Replacement, Reissue, Revocation & Recovery	Credential lifecycle changes
15.3.7	Failed Generation, Delivery & Credential Exception Management	Operational exceptions
15.3.	Credential Usage & Cross-Media Traceability	Multi-media usage trace

Screen	Backend Screen	Primary Responsibility
8		
15.3.9	Credential Security, Audit & Operational Evidence	Security/audit
15.3.10	Ticket Media Analytics & AI Operations Intelligence	Analytics, prediction & AI

Area 15 — Complete Architecture

With Board 3, **Area 15** is now fully defined as **3 Boards / 30 backend screens**.

Board 1 — Define the Identity

Virtual Ticket → Lifecycle → Media Registry → Multi-Media Binding → Credential Resolver → Shared Entitlement → Activation/Fallback → Replacement Rules

Board 2 — Design the Media

Select Media → Design → Dynamic Fields → Wallet/QR/RFID Configuration → Branding → Localization → Preview → Test → Publish

Board 3 — Operate the Credentials

Generate → Bind → Deliver → Activate → Monitor → Use → Replace/Revoke → Resolve Exceptions → Audit → Optimize

The most important Area 15 rule

We recommend sees this architecture diagram prominently in the SOW:

ONE VIRTUAL TICKET

- ✓ **QR / Dynamic QR**
- ✓ **Apple Wallet**
- ✓ **Google Wallet**
- ✓ **RFID / NFC**
- ✓ **Card / Wristband**
- ✓ **Face Recognition Reference**
- ✓ **Future Media**

All resolve to the same authoritative ticket, entitlement and usage state.

This means that if the customer enters using **face recognition**, TICVAI does not consume a “face ticket.” It resolves the face credential to the **Virtual Ticket**, and the Virtual Ticket's entitlement is evaluated/updated. Presenting QR or RFID afterward still resolves to that same ticket.

That principle should remain non-negotiable when we move next into **Area 16 — Access Control & Dynamic Credential Security**, because Area 16 should consume this Virtual Ticket/Credential Resolver architecture rather than inventing another credential model.